

[Page 1] Introduction & Learning Objectives

1. Lesson Overview

The flipped classroom model places new instruction at home through video lectures and reserves in-class time for exercises, projects, and discussions. Research indicates that students who engage actively with material through collaborative problem-solving demonstrate significantly higher retention rates compared to passive listening. The teacher's role shifts from lecturer to facilitator, providing targeted assistance while students work through higher-order thinking tasks.

2. Real-Time Sync Protocol

Real-time whiteboard synchronisation relies on an event-driven WebSocket protocol. When a teacher moves a page index, the SDK emits a scene-change event that propagates to all participants in the room within milliseconds. Participants poll the shared scene state at 1-second intervals to handle reconnection scenarios. A PDF document uploaded before session start is converted to a series of whiteboard scenes, each corresponding to one page, enabling seamless teacher-controlled navigation.

3. Session Notes

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[Page 2] Core Concepts — Part 1

1. Conceptual Framework

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2. Pedagogical Approach

The flipped classroom model places new instruction at home through video lectures and reserves in-class time for exercises, projects, and discussions. Research indicates that students who engage actively with material through collaborative problem-solving demonstrate significantly higher retention rates compared to passive listening. The teacher's role shifts from lecturer to facilitator, providing targeted assistance while students work through higher-order thinking tasks.

3. Technical Background

Real-time whiteboard synchronisation relies on an event-driven WebSocket protocol. When a teacher moves a page index, the SDK emits a scene-change event that propagates to all participants in the room within milliseconds. Participants poll the shared scene state at 1-second intervals to handle reconnection scenarios. A PDF document uploaded before session start is converted to a series of whiteboard scenes, each corresponding to one page, enabling seamless teacher-controlled navigation.

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[Page 3] Core Concepts — Part 2

1. Deep Dive: WebSocket Events

Real-time whiteboard synchronisation relies on an event-driven WebSocket protocol. When a teacher moves a page index, the SDK emits a scene-change event that propagates to all participants in the room within milliseconds. Participants poll the shared scene state at 1-second intervals to handle reconnection scenarios. A PDF document uploaded before session start is converted to a series of whiteboard scenes, each corresponding to one page, enabling seamless teacher-controlled navigation.

Each classroom session is identified by a unique UUID derived from the course ID and timestamp. Session state is persisted in DynamoDB, ensuring that concurrent classroom sessions — even those running on the same server — remain fully isolated. No participant state, whiteboard content, or PDF attachment from one session can bleed into another.

3. Further Reading

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[Page 4] Practice Exercises

1. Exercise A — Concept Map

Draw a concept map connecting the following terms: WebSocket, DynamoDB, Scene Index, PDF Upload, Session Key, Broadcast Channel. For each connection, annotate the direction and nature of the data flow. Use the whiteboard tools to draw and annotate your map directly on this page.

2. Exercise B — Discussion Questions

1. Why is it important to use a sequential ready-click pattern rather than a parallel one when multiple groups are joining simultaneously?
2. How does the DynamoDB write model prevent race conditions during high-concurrency classroom entry?
3. What fallback mechanisms exist when the Agora RTM channel is unavailable?

3. Group Activity

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[Page 5] Summary & Assessment

1. Key Takeaways

After completing this module, students should be able to: (1) explain the end-to-end flow of a classroom PDF sync event; (2) describe how session isolation is achieved using unique UUID keys; (3) identify potential failure points and apply the appropriate diagnostic sequence; (4) configure SLO thresholds for whiteboard sync latency in a production environment.

2. Self-Assessment Checklist

- %j I can explain the role of sessionReadyKey in PDF metadata s
- %j I understand why PDF upload must happen before entering th
- %j I can read the Agora whiteboard scene state from the browse
- %j I know how to use the load-escalation test suite to verify cor

3. References

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